

TokTickIT

CPE334 Lab 3 - Release Evidence

Nine-part submission covering workflow, specification, traceability, AI use, authentication, Staff operations, Administration, and responsive visual evidence.

Field	Release-candidate value
Repository	MacOverlorD/toktickit
Integration baseline	lab3-staging @ 93d270d
Submission branch	feature/3-09-release-submission
Verification date	2026-09-16 (Asia/Bangkok)
Current gate	Peer review and final-main verification remain required

Integrity note

This PDF reports the GitHub record as it exists. PR #50 has a positive follow-up comment and reviewer merge, but no later formal approval. AI assistance is disclosed in the AI-use section. The final-main result is not claimed before the release PR is reviewed and merged.

Answer Part 1 - Git and Workflow

Lab 3 was delivered as nine scoped issues on a GitHub Project. Feature branches started from lab3-staging, targeted that branch through pull requests, and preserved main for the reviewed release.

Issue	Scope	PR / result
#33	Engineering contract	PR #42 - approved and merged
#34	Data migration	PR #44 - fixes merged
#35	Authentication and shell	PR #45 - fixes merged
#36	Requester regression	PR #46 - fixes merged
#37	Staff queue	PR #47 - fixes merged
#38	Detail/workflow/communication	PR #48 - fixes merged
#39	User management	PR #49 - fixes merged
#40	Integrated verification	PR #50 - reviewer merged
#41	Release and evidence	This release-candidate branch

Review evidence

- The exact collaborator identity, comments, fixes, commits, and decision state are recorded in [docs/lab-03/reviewer.md](#).
- PR #42 has a formal approval. PRs #44-#50 were merged after fixes, but no later formal approval is invented where GitHub still records Changes requested.
- Repository structure documents client, server, Lab 3 browser tests, reviewed screenshots, and the single output PDF. Generated reports and temporary results remain ignored.

Answer Part 2 - Specification-Driven Development

The specification was merged before feature implementation. It defines the contract for authentication, authorization, eight-state ticket workflow, communication, migration, responsive UI, and submission evidence.

Artifact	Evidence
Functional requirements	FR-01 through FR-10 (10 total)
Business rules	BR-01 through BR-26 (26 total)
Acceptance criteria	AC-01 through AC-17 (17 total)
Design decisions	D-01 through D-07 with rationale
Baseline PR	PR #42

Traceable implementation choices

- Server-owned identity replaced spoofable requester headers and client identity selectors.
- Opaque HttpOnly sessions, exact Origin and CSRF checks, Argon2id password verification, session rotation, and concurrent-safe login limits implement the security contract.
- Ticket operations use allowlisted transitions, role checks, current-version preconditions, transactions, and safe conflict feedback.
- Migration preserves Lab 2 data and relationships, preflights legacy email validity, and keeps seed identity independent from editable email.

Primary artifact: <docs/lab-03/specification.md>

Answer Part 3 - Test-Driven Development and Traceability

Every acceptance criterion is mapped to API, UI, unit, migration, browser, accessibility, or release verification. Regression tests were added with each reviewer finding and defect.

Release-candidate command	Result
npm run prisma:generate --prefix server	Passed
npm run test:server	24 files / 144 tests passed
npm run test:client	17 files / 134 tests passed
npm run build	Server/client TypeScript and Vite passed
npm run test:e2e	10 Playwright tests passed in 2.3 min

Defect-to-regression example

A development API already running on port 3000 exposed an unmocked unit-test request. Its real 401 emitted the application's unauthenticated event and redirected the test to Login. The shared client test setup now rejects unexpected fetches, while the inherited Attachment suite explicitly mocks its communication dependency. The full client suite was rerun and passed.

Release gate

REL-01 is **In progress**: the complete release-candidate suite passes, but the same verification must run again on final main after peer review and merge. This PDF does not claim that future result.

Full matrix and per-issue commands: [docs/lab-03/tests.md](https://docs.tockit.com/lab-03/tests.md)

Answer Part 4 - AI Use and Reflection

OpenAI Codex was used for repository inspection, specification drafting, implementation, test generation, review-fix work, verification, evidence organization, and this disclosed PDF. Selected prompts and an issue-by-issue work record are retained in the repository.

My Reflection

AI was most useful for turning the Lab 3 handout into traceable requirements, implementing repetitive API/UI tests, and checking the same behavior across roles and screen sizes. I did not treat generated work or self-review as proof: I used executable server, client, build and browser checks, inspected visual evidence, and kept the collaborator's formal review state separate from positive comments. The main limitation was environment sensitivity, demonstrated when a running development API changed a unit-test result; isolating network access and rerunning the complete suites was necessary before trusting the evidence.

Student confirmation

This reflection is a Codex-assisted draft based on the recorded work. The student must confirm that it matches their experience before submission.

Disclosure log: <docs/lab-03/ai-use.md>

Answer Part 5 - Login and Password Change UI

The login flow provides generic failure feedback, rate-limit handling, session restoration, role-aware routing, logout, and mandatory first-password change. Protected routes cannot be reached by changing a URL or client state.

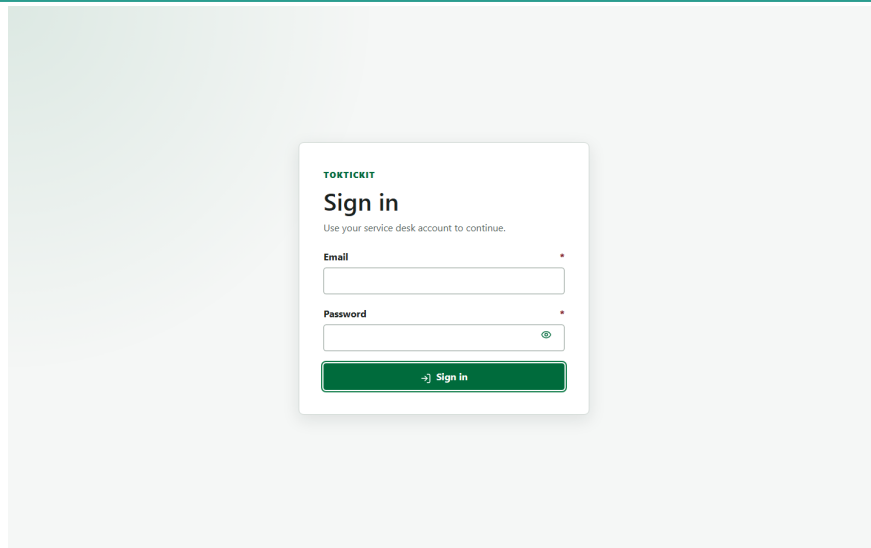


Figure: Desktop login with explicit labels and focused, keyboard-reachable controls - artifacts/lab-03/screenshots/authentication/login-desktop.png

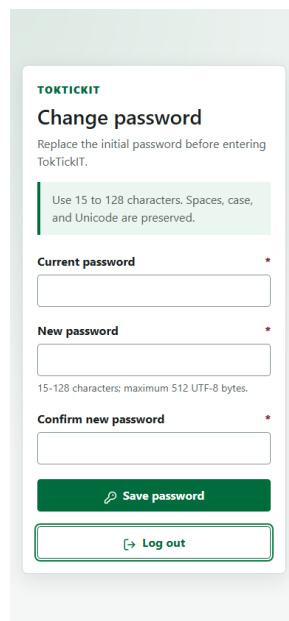
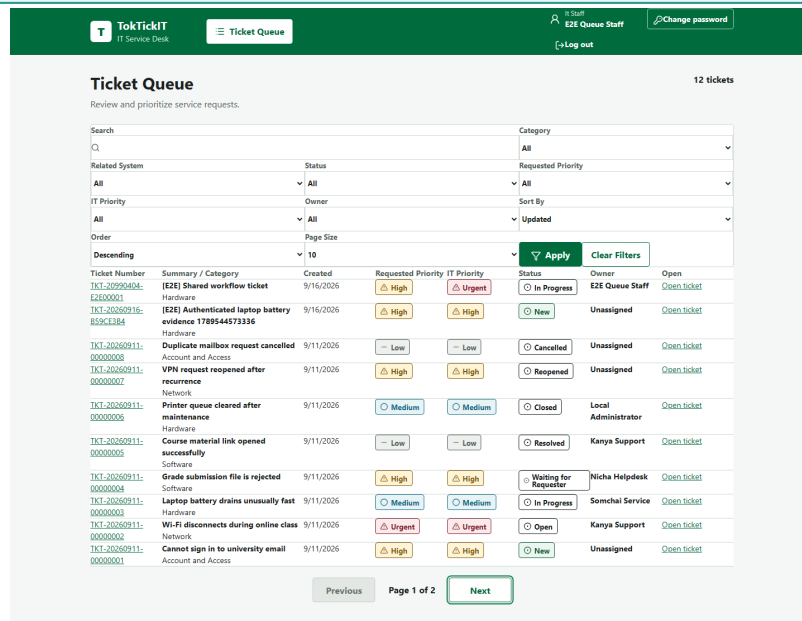


Figure: Mandatory password change at mobile width - artifacts/lab-03/screenshots/authentication/mandatory-change-mobile.png

Answer Part 6 - IT Staff Ticket Queue

Authorized IT Staff can search, filter, sort, page, and change page size. Controls are URL-backed and immediate; the desktop table and compact mobile cards preserve ticket, requester, status, priority, and assignment context.



Ticket Number	Summary / Category	Created	Requested Priority	IT Priority	Status	Owner	Open
TXT-20250911-00000001	[E2E] Shared workflow ticket	9/16/2025	High	Urgent	In Progress	E2E Queue Staff	Open ticket
TXT-20250911-00000002	Hardware	9/16/2025	High	High	New	Unassigned	Open ticket
TXT-20250911-00000003	Duplicate mailbox request cancelled	9/11/2025	Low	Low	Cancelled	Unassigned	Open ticket
TXT-20250911-00000004	Account and Access	9/11/2025	High	High	Reopened	Unassigned	Open ticket
TXT-20250911-00000005	VPN request reopened after recurrence	9/11/2025	Medium	Medium	Closed	Local Administrator	Open ticket
TXT-20250911-00000006	Printer queue cleared after maintenance	9/11/2025	Low	Low	Resolved	Kanya Support	Open ticket
TXT-20250911-00000007	Course material link opened successfully	9/11/2025	High	High	Waiting for Requester	Nicha Helpdesk	Open ticket
TXT-20250911-00000008	Software	9/11/2025	Medium	Medium	In Progress	Somchal Service	Open ticket
TXT-20250911-00000009	Laptop battery drains unusually fast	9/11/2025	Urgent	Urgent	Open	Kanya Support	Open ticket
TXT-20250911-00000010	Wi-Fi disconnects during online class	9/11/2025	High	High	New	Unassigned	Open ticket
TXT-20250911-00000011	Cannot sign in to university email	9/11/2025	High	High	New	Unassigned	Open ticket
TXT-20250911-00000012	Account and Access	9/11/2025	High	High	New	Unassigned	Open ticket

Figure: Eight-column Staff Queue desktop table - artifacts/lab-03/screenshots/staff-queue/queue-desktop.png



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TXT-20250911-00000004	Account and Access	9/11/2025	High	High	Reopened	Unassigned	Open ticket
TXT-20250911-00000005	VPN request reopened after recurrence	9/11/2025	Medium	Medium	Closed	Local Administrator	Open ticket
TXT-20250911-00000006	Printer queue cleared after maintenance	9/11/2025	Low	Low	Resolved	Kanya Support	Open ticket
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TXT-20250911-00000011	Cannot sign in to university email	9/11/2025	High	High	New	Unassigned	Open ticket
TXT-20250911-00000012	Account and Access	9/11/2025	High	High	New	Unassigned	Open ticket

Figure: Staff Queue mobile cards and navigation - artifacts/lab-03/screenshots/staff-queue/queue-mobile.png

Answer Part 7 - Staff Ticket Detail, Workflow, and Communication

The Staff detail view combines operational context, read-only attachments, assignment, priority, exact status transitions, optimistic versions, Public Comments, and private Internal Notes. Transactions recheck the current actor and return explicit stale-conflict recovery.

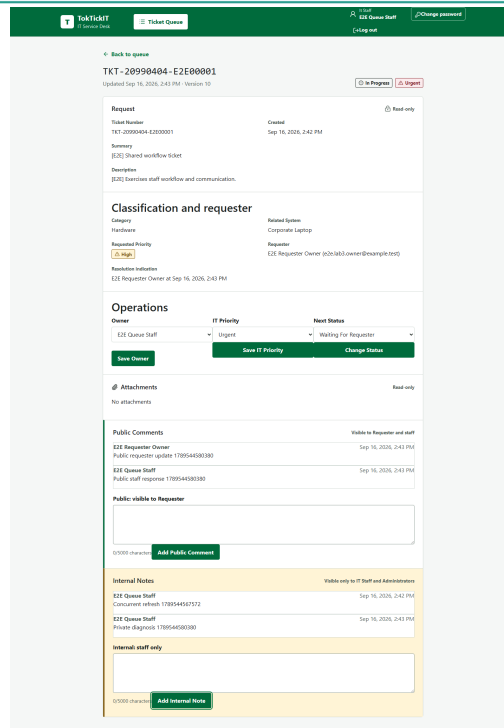


Figure: Staff Ticket Detail with workflow and communication - artifacts/lab-03/screenshots/staff-ticket-detail/ticket-detail-desktop.png

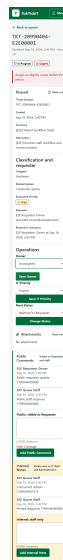


Figure: First-invalid operation focus and readable feedback at 320px - artifacts/lab-03/screenshots/staff-ticket-detail/operation-validation-boundary-320.png

Answer Part 8 - Administrator User Management

Administrators can search/filter accounts and create, edit, activate/deactivate, change roles, or reset passwords. Canonical email rules, optimistic versions, session invalidation, current-actor checks, serializable last-admin protection, and atomic ticket-owner cleanup protect the workflow.

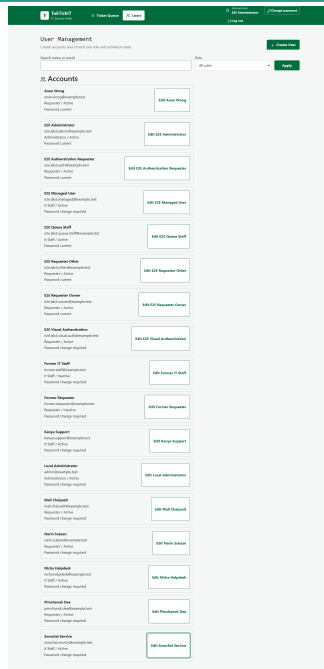


Figure: Administrator account list and editor at desktop width - artifacts/lab-03/screenshots/user-management/accounts-desktop.png

Somchai Service

somchai.service@example.test

It Staff / Active

Password change required

Edit Somchai Service

Create User

Name

Name must contain 1 to 120 valid characters.

Email

Enter a valid email address.

Role

REQUESTER

☒ Active account

Initial password

Password must contain 15 to 128 characters and no more than 512 UTF-8 bytes.

Confirm password

Figure: Create-user validation and first-invalid focus at 320px - artifacts/lab-03/screenshots/user-management/create-user-validation-boundary-320.png

Answer Part 9 - Zen Green UI and Responsive Verification

The shared Zen Green visual system uses calm teal/green surfaces, strong text contrast, consistent spacing, role-aware navigation, visible keyboard focus, flat data presentation, and responsive layouts verified at desktop, tablet, mobile, and the 320px boundary.

Evidence area	Coverage
Authentication	Desktop, tablet, mobile, boundary, validation
Requester	List, detail, comments, four widths
Staff Queue	Table/cards, filters, navigation, four widths
Staff Detail	Operations, communication, validation, four widths
User Management	List/editor, validation, four widths
Keyboard	Complete target traversal, visible focus, mobile navigation

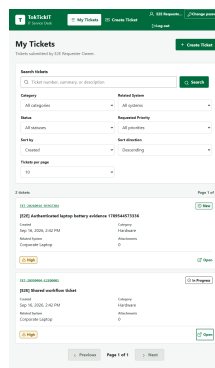


Figure: Zen Green Requester list at tablet width - artifacts/lab-03/screenshots/requester/my-tickets-tablet.png



Figure: Role-aware Administrator navigation and cards at mobile width - artifacts/lab-03/screenshots/user-management/accounts-mobile.png

Submission checkpoint

This release-candidate document contains all nine required answer headings and links its claims to tracked source evidence. Peer review, release merge, and final-main rerun remain explicit gates before Issue #41 can be marked Done.